

**Institut Universitaire de Technologie,
Aix-Marseille Université**

ANNEXES
RAPPORT DE STAGE de fin de deuxième année
Bachelor Universitaire de Technologie
Spécialité Réseaux et Télécommunications
parcours cybersécurité

Renforcement de la Sécurité Informatique et
Optimisation des Services IT

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CAMPUS PROVENCE VERTE

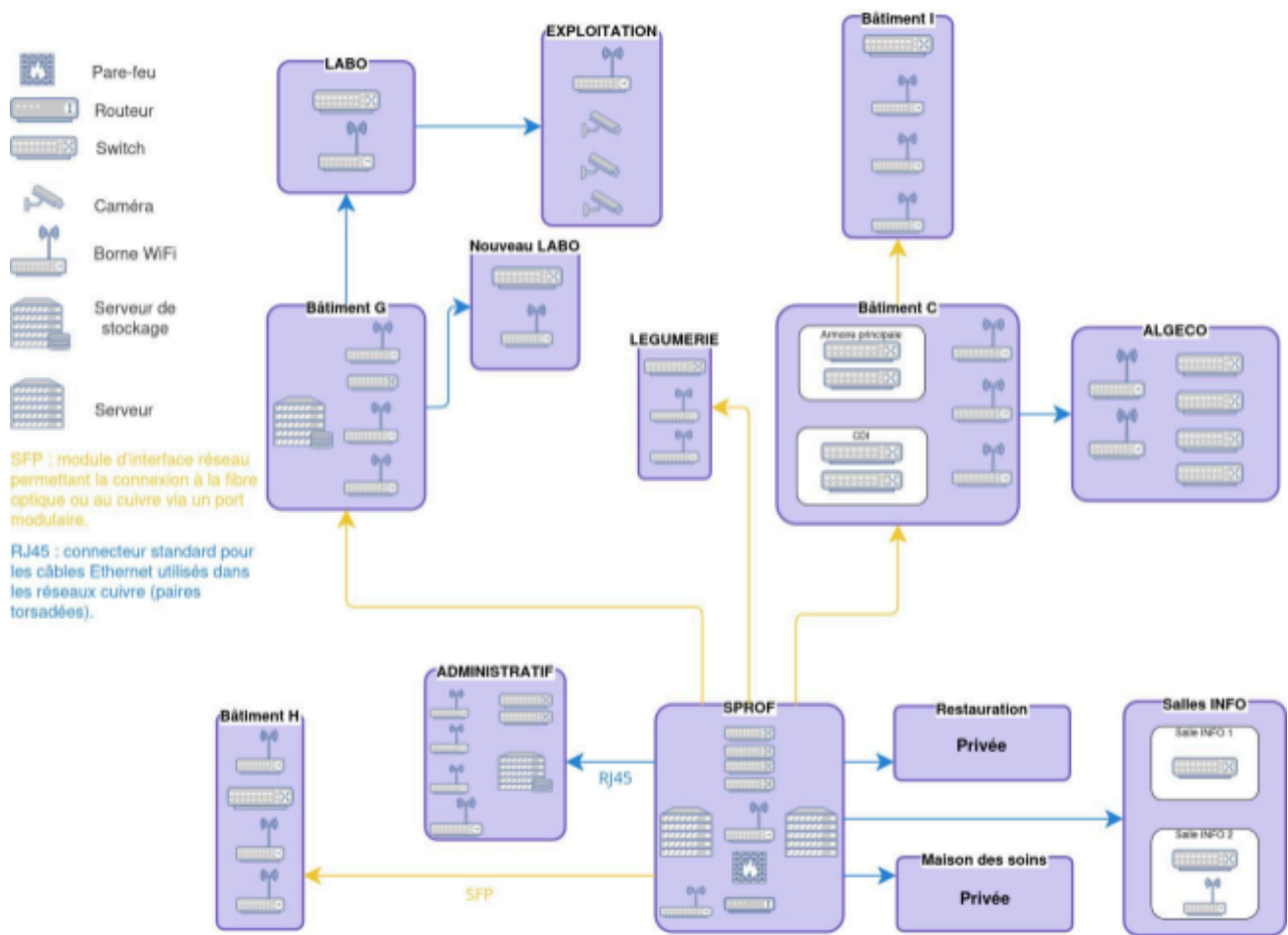
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Schéma réseau	5
Liste des composants principaux et configuration du serveur Proxmox (Dell PowerEdge T640)	5
Résultat de la commande netstat laputen sur SE4AD	6
Résultat de la commande systemctl list-units --type=service --state=running sur SE4AD	6
Résultat de la commande systemctl list-units --type=service --state=running sur SE4FS	7
fixssh.sh	7
secure-upgrade.sh	8
.bashrc	8

Schéma réseau



Liste des composants principaux et configuration du serveur Proxmox (Dell PowerEdge T640)

- Intel(R) Xeon(R) Silver 4110 CPU @ 2.10Ghz x2 (une allocation de CPU a été faite: le premier CPU pour SE4FS, et deuxième CPU est partagé entre SE4AD et le DNS)
- 32 Go RAM DDR4 (26 Go pour le SE4FS, 4 pour le DNS, 2 pour SE4AD)
- 2 Contrôleurs Ethernet réseaux 10 G
- 4 Contrôleurs Ethernet réseaux Gigabit
- Carte mère PowerEdge T640 (Figure 1)
- Contrôleurs de stockage:
 - Internes: PERC H330, H730P, H740P, HBA330
 - Externe HBA (RAID): PERC H840
 - Externe HBA (NON-RAID): SAS 12 Gbit/s

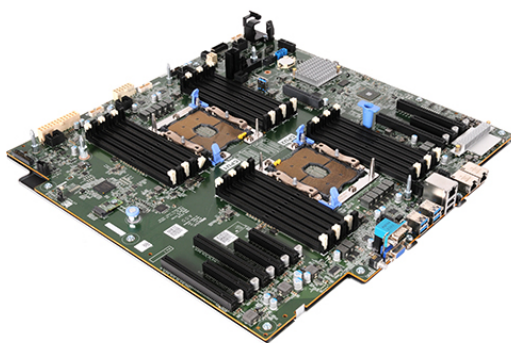


Figure 1: Carte mère

Résultat de la commande netstat laputen sur SE4AD

tcp	0	0 0.0.0.0:53	0.0.0.0:*	LISTEN	0	11752	400/samba: task[dns
tcp	0	0 0.0.0.0:22	0.0.0.0:*	LISTEN	0	11488	357/sshd: /usr/sbin
tcp	0	0 0.0.0.0:88	0.0.0.0:*	LISTEN	0	11615	376/samba: task[kdc
tcp	0	0 127.0.0.1:6010	0.0.0.0:*	LISTEN	0	16864	904/sshd: root@pts/
tcp	0	0 126.10.3.201:3268	126.10.3.62:49927	TIME_WAIT	0	0	-
tcp	0	320 126.10.3.201:22	126.10.11.167:50326	ESTABLISHED	0	16803	904/sshd: root@pts/
tcp6	0	0 :::636	:::*	LISTEN	0	11805	372/samba: task[lda
tcp6	0	0 :::445	:::*	LISTEN	0	11903	371/smbd
tcp6	0	0 :::49152	:::*	LISTEN	0	11586	364/samba: task[rpc
tcp6	0	0 :::49153	:::*	LISTEN	0	11687	392/samba: task[rpc
tcp6	0	0 :::49154	:::*	LISTEN	0	11699	392/samba: task[rpc
tcp6	0	0 :::3268	:::*	LISTEN	0	11806	372/samba: task[lda
tcp6	0	0 :::3269	:::*	LISTEN	0	11807	372/samba: task[lda
tcp6	0	0 :::389	:::*	LISTEN	0	11804	372/samba: task[lda
tcp6	0	0 :::135	:::*	LISTEN	0	11703	392/samba: task[rpc
tcp6	0	0 :::139	:::*	LISTEN	0	11904	371/smbd
tcp6	0	0 :::464	:::*	LISTEN	0	11613	376/samba: task[kdc
tcp6	0	0 :::53	:::*	LISTEN	0	11750	400/samba: task[dns
tcp6	0	0 :::22	:::*	LISTEN	0	11499	357/sshd: /usr/sbin
tcp6	0	0 :::88	:::*	LISTEN	0	11611	376/samba: task[kdc
tcp6	0	0 ::1:6010	:::*	LISTEN	0	16863	904/sshd: root@pts/
udp	0	0 126.10.3.201:123	0.0.0.0:*		0	11378	353/ntpd
udp	0	0 127.0.0.1:123	0.0.0.0:*		0	11376	353/ntpd
udp	0	0 0.0.0.0:123	0.0.0.0:*		0	11372	353/ntpd
udp	0	0 126.10.3.201:137	0.0.0.0:*		0	11552	366/samba: task[nbt
udp	0	0 126.255.255.255:137	0.0.0.0:*		0	11551	366/samba: task[nbt
udp	0	0 0.0.0.0:137	0.0.0.0:*		0	11549	366/samba: task[nbt
udp	0	0 126.10.3.201:138	0.0.0.0:*		0	11554	366/samba: task[nbt
udp	0	0 126.255.255.255:138	0.0.0.0:*		0	11553	366/samba: task[nbt
udp	0	0 0.0.0.0:138	0.0.0.0:*		0	11550	366/samba: task[nbt
udp	0	0 126.10.3.201:389	0.0.0.0:*		0	11658	374/samba: task[cld
udp	0	0 0.0.0.0:389	0.0.0.0:*		0	11657	374/samba: task[cld
udp	0	0 126.10.3.201:464	0.0.0.0:*		0	11620	376/samba: task[kdc
udp	0	0 0.0.0.0:464	0.0.0.0:*		0	11618	376/samba: task[kdc
udp	0	0 0.0.0.0:53	0.0.0.0:*		0	11753	400/samba: task[dns
udp	0	0 126.10.3.201:88	0.0.0.0:*		0	11619	376/samba: task[kdc
udp	0	0 0.0.0.0:88	0.0.0.0:*		0	11616	376/samba: task[kdc
udp6	0	0 fe80::b86a:6dff:fe8:123	:::*		106	11954	353/ntpd
udp6	0	0 ::1:123	:::*		0	11380	353/ntpd
udp6	0	0 :::123	:::*		0	11369	353/ntpd
udp6	0	0 :::389	:::*		0	11656	374/samba: task[cld
udp6	0	0 ::1:464	:::*		0	11614	376/samba: task[kdc
udp6	0	0 :::53	:::*		0	11751	400/samba: task[dns
udp6	0	0 :::88	:::*		0	11612	376/samba: task[kdc
udp	0	0 0.0.0.0:138	0.0.0.0:*		0	11550	366/samba: task[nbt
udp	0	0 126.10.3.201:389	0.0.0.0:*		0	11658	374/samba: task[cld

Résultat de la commande systemctl list-units --type=service --state=running sur SE4AD

```
root@se4ad:~# systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
getty@tty1.service	loaded	active	running	Getty on tty1
ntp.service	loaded	active	running	Network Time Service
rsyslog.service	loaded	active	running	System Logging Service
samba-ad-dc.service	loaded	active	running	Samba AD Daemon
ssh.service	loaded	active	running	OpenBSD Secure Shell server
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-udev.service	loaded	active	running	Rule-based Manager for Device Events and File
user@0.service	loaded	active	running	User Manager for UID 0

11 loaded units listed.

LOAD = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB = The low-level unit activation state, values depend on unit type.

Résultat de la commande `systemctl list-units --type=service --state=running` sur SE4FS

```
root@se4fs:~# systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
apache2.service	loaded	active	running	The Apache HTTP Server
atftpd.service	loaded	active	running	LSB: Launch atftpd server
avahi-daemon.service	loaded	active	running	Avahi mDNS/DNS-SD Stack
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
cron.service	loaded	active	running	Regular background program processing daemon
cups-browsed.service	loaded	active	running	Make remote CUPS printers available locally
cups.service	loaded	active	running	CUPS Scheduler
dbus.service	loaded	active	running	D-Bus System Message Bus
freeradius.service	loaded	active	running	FreeRADIUS multi-protocol policy server
getty@tty1.service	loaded	active	running	Getty on tty1
isc-dhcp-server.service	loaded	active	running	LSB: DHCP server
mariadb.service	loaded	active	running	MariaDB 10.5.15 database server
nfs-blkmap.service	loaded	active	running	pNFS block layout mapping daemon
nfs-idmapd.service	loaded	active	running	NFSv4 ID-name mapping service
nfs-mountd.service	loaded	active	running	NFS Mount Daemon
nmdbd.service	loaded	active	running	Samba NMB Daemon
php7.4-fpm.service	loaded	active	running	The PHP 7.4 FastCGI Process Manager
rpcbind.service	loaded	active	running	RPC bind portmap service
rsyslog.service	loaded	active	running	System Logging Service
smbd.service	loaded	active	running	Samba SMB Daemon
ssh.service	loaded	active	running	OpenBSD Secure Shell server
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-timesyncd.service	loaded	active	running	Network Time Synchronization
systemd-udevd.service	loaded	active	running	Rule-based Manager for Device Events and Files
user@.service	loaded	active	running	User Manager for UID 0
winbind.service	loaded	active	running	Samba Winbind Daemon
xinetd.service	loaded	active	running	LSB: Starts or stops the xinetd daemon.

LOAD = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB = The low-level unit activation state, values depend on unit type.
28 loaded units listed.

fixssh.sh

```
#!/bin/bash
file="/etc/ssh/sshd_config"
backup="$file.bak"
port=""
# Parse les options
while [[ $# -gt 0 ]]; do
    case "$1" in
        -p|--port)
            port="$2"
            shift 2
            ;;
        *)
            echo "Usage: $0 [-p PORT]"
            exit 1
            ;;
    esac
done
# Si aucun port fourni en argument, demande-le à l'utilisateur
if [ -z "$port" ]; then
    read -p ">>> Nouveau port SSH: " port
fi
# Vérifie que le port est bien un nombre
```

```

if ! [[ "$port" =~ ^[0-9]+$ ]]; then
    echo "Erreur: Le port doit être un nombre."
    exit 1
fi
# Sauvegarde du fichier
cp "$file" "$backup"
echo "Sauvegarde créée: $backup"
# Modifie la ligne du port (ou l'ajoute si elle n'existe pas)
if grep -q "^Port" "$file"; then
    sed -i "s/^Port .*/Port $port/" "$file"
else
    echo "Port $port" >> "$file"
fi
echo "Port SSH mis à jour à $port"

```

secure-upgrade.sh

```

#!/bin/bash
SE4AD_IP="126.10.3.201"
PORT_INITIAL=6602
PORT_TEMP=22
SSH_USER="root"
echo "Autorisation temporaire du port $PORT_TEMP sur SE4AD"
ssh -p $PORT_INITIAL $SSH_USER@$SE4AD_IP "
    ufw allow $PORT_TEMP &&
    ufw reload &&
    fixssh -p $PORT_TEMP &&
    systemctl restart ssh
"
echo "Passage sur le port temporaire $PORT_TEMP"
sleep 5
echo "Mise à jour sur SE4FS"
apt update -y && apt upgrade -y
echo "Restauration de l'ancien port sur SE4AD"
ssh -p $PORT_TEMP $SSH_USER@$SE4AD_IP "
    ufw allow $PORT_INITIAL &&
    fixssh -p $PORT_INITIAL &&
    ufw delete allow $PORT_TEMP &&
    ufw reload &&
    systemctl restart ssh
"
echo "Opération terminée avec succès."

```

.bashrc

```

# ~/.bashrc: executed by bash(1) for non-login shells.
export MANPATH="$HOME/.local/share/man:$MANPATH"
# Note: PS1 and umask are already set in /etc/profile. You should not
# need this unless you want different defaults for root.
# PS1='${debian_chroot:+($debian_chroot)}\h:\w\$ '
# umask 022
# You may uncomment the following lines if you want `ls` to be colorized:

```

```

export LS_OPTIONS='--color=auto'
eval "$(dircolors)"
alias ls='ls $LS_OPTIONS'
alias ll='ls $LS_OPTIONS -l'
alias l='ls $LS_OPTIONS -lA'
# Some more alias to avoid making mistakes:
alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'
# Commandes personnalisées
alias fixssh='./fixssh.sh'
alias servicereading='systemctl list-units --type=service --state=running'
alias grep='grep --color=auto'
alias ip='ip -c a'
alias listen='lsof -i -P -n | grep LISTEN'
apt () {
    if [[ "$1" == "upgrade" && "$2" == "-SSH" ]]; then
        ./secure-upgrade.sh
    else
        command apt "$@"
    fi
}

```